

PROFESSIONAL SUMMARY

I am a passionate Ph.D. candidate in Forensic Science with a background in Plant Biology and Plant Biotechnology. Expertise in forensic science techniques for crime scene investigation and analysis. Proven ability to prioritize tasks, solve problems, and manage complex projects. Driven by a thirst for knowledge and a commitment to Mental health, I lead an interdisciplinary Ph.D. project titled "Exploring Mental Health Status in a Selected Population: A corpus analysis combining Forensic Linguistics and Psychology - a Mixed Method Study." This project utilizes forensic linguistics, psychology, and machine learning/natural language processing (NLP) to develop innovative methods for analyzing the Mental Health Status of an individual. Eager to leverage my research and foster international collaboration, I am committed to contributing to a safer and more just world.

EDUCATIONAL BACKGROUND

Ph.D. Candidate in Forensic Science (Thesis Submission by December, 2026)

Kasturba Medical College, Manipal (2023–Present)

Leading an interdisciplinary Ph.D. project applying forensic linguistics, psychology, and machine learning to understand an early detection of Mental Illness in Individuals.

Master of Science in Forensic Science (2022)

Kasturba Medical College, Manipal (2020–2022)

Graduated with a CGPA of 8.38, demonstrating a solid foundation in forensic science principles and techniques. As a critical team member, I have contributed to complex projects such as crime scene simulations, forensic analyses, and research initiatives.

Bachelor of Science in Plant Biology and Plant Biotechnology (2020)

Loyola College, Chennai (2017–2020)

Graduated with a CGPA of 7.92, demonstrating a solid foundation in plant science and plant biotechnology. Possess expertise in advanced instrumentation and techniques acquired through a specialization in Microbial & Environmental Biotechnology, Genetics, and Plant Breeding.

Secondary (SSLC) and Higher Secondary (HSC) (2015, 2017)

SRVS National Higher Secondary School, Karaikal (2014 – 2017)

Completed Higher Secondary Education and Secondary Education from the SRVS National Higher Secondary School, Karaikal, with 74.00% in May 2017 and 97.00% in May 2015, respectively.

RESEARCH EXPERIENCE

ORCID ID: [0000-0002-7824-2881](#)

SCOPUS ID: [59896201000](#)

Primary Research Focus in PhD

This study aims to improve how we understand mental health through written text. Traditionally, researchers relied on surveys and interviews, but these methods have limitations. Existing text analysis techniques have not yet considered all relevant aspects, like demographics and psychological factors, and have not been applied across diverse populations. To address this, the study proposes using Machine Learning (ML) and Natural Language Processing (NLP) on written text data. This builds upon traditional methods and explores a more efficient approach. By comparing manual analysis with ML/NLP, the study hopes to demonstrate the effectiveness of these techniques in understanding mental health, particularly for students, Ph.D. scholars, and LGBT individuals.

PG Research Work

Title: “Comparison of Genuine Suicide Note and Simulated Suicide Note based on Forensic Linguistic Methods.”

Completed a dissertation under the guidance of **Dr. Shankar N Bakkannavar**.

Description:

In recent years, there have been several more factors that contribute to suicide. In this situation, few people write a note/letter before they commit suicide. These writings are said to be suicide notes, which explain the reason(s) for their decision to commit suicide. They use this note to communicate ideas and sentiments that the individual could not express daily. Also, sometimes, they write a note to provide instructions for the disposition of the remains. Occasionally, to confess to a murder or other crimes. In this study, the main aim is to compare and analyze the genuine suicide notes and simulated suicide notes written by both men and women based on forensic linguistic methods. Several features are found in the suicide notes based on the structure, content, and many other factors. In this, we have achieved the objective of exploring features found in a genuine and simulated suicide note by undergoing qualitative descriptive analysis and comparing and identifying the commonalities and differences in the content and format of the notes.

UG Research Work

Title: “Impacts of Botanical Pesticides on *Allium cepa*.”

Completed a research park project under the guidance of **Dr. R. Ravindhran**.

Description:

The morphological and cytological effects of water extracts of three different medicinal plants, Viz, *Lantana camara*, *Aerva lanata*, and *Vitex negundo*, were studied in *Allium cepa* L. Plant material was collected from different regions and was processed to prepare water extracts. Different concentrations of water extract (5%, 10%, 15%, 20%, 25%) were prepared and applied under in situ and ex-situ conditions. The treated root samples showed varied

morphological and cytological features. The present study indicates the cytotoxic properties of three different medicinal plants.

WORK EXPERIENCE

Internship Program on Questioned Document Division at Central Forensic Science Laboratory (CFSL), Bhopal.

DURATION: 1 MONTH

Description:

My internship at the Central Forensic Science Laboratory's (CFSL) Questioned Document Section provided invaluable experience in forensic document examination. I worked on genuine cases submitted by various Central agencies, including the Central Bureau of Investigation (CBI) and the Anti-Corruption Bureau (ACB). The CFSL's approach to knowledge transfer and mentorship is unparalleled. I am particularly grateful to **Sh. Rajat Choudhary**, Scientist-B(Document), for his guidance and supervision throughout the internship.

Short Internship Training at Telangana State Forensic Science Laboratory, Hyderabad.

DURATION: 1 WEEK

Description:

I was privileged to participate in a one-week intensive training program at the Telangana State Forensic Science Laboratory (TS FSL). This valuable experience exposed me to a wide range of the laboratory's divisions, including their impressive and fully functional Cyber Forensic Lab actively handling live cases. I sincerely thank **Mrs. Dr. Anitha Evangelin** for her exceptional mentorship throughout the training. Her guidance and expertise provided me with invaluable insights and experiences.

Internship Training in Kottucherry Police Station, Karaikal District, Puducherry, UT.

DURATION: 1 WEEK

Description:

During my internship at the Kottucherry Police Station, I was entrusted with the responsibilities of an intern. This valuable experience provided me with firsthand insights into police station protocols and the diverse duties of the officers there. I sincerely thank **Shri R. Lokeshwaran**, I.P.S., Senior Superintendent of Police (SSP) of the Karaikal District, for his exceptional mentorship throughout the internship. His guidance proved invaluable in my learning and professional development.

Internship in Cyber Cell at CB CID Headquarters, Egmore, Chennai.

DURATION: 1 WEEK

Description:

I had the opportunity to collaborate with the CB-CID-CYBER CELL team. Through this collaboration, I gained familiarity with the tools and technologies they utilize to facilitate cybercrime investigations. This experience also provided valuable insights into optimizing investigative procedures to maximize the success of police officers in solving cybercrime cases.

Internship Program with Legal Desire Media and Insights.

DURATION: 1 MONTH

Description:

During the early stages of my master's program, I actively sought internship opportunities and identified a relevant position advertised on Instagram by Legal Desire. Following a competitive application process, I was selected from over 400 candidates for a one-month online internship. This experience marked my first foray into online internships and proved to be a valuable learning opportunity. I am grateful to **Mr. Anuj Kumar** for extending this opportunity and to **Ms. Kimee Minj** for providing insightful guidance throughout the training program.

PUBLICATION

Journal : Scientific Reports (Impact Factor : **4.9**) (**Q1** – 87th Percentile)

Mani, R.T., Palimar, V., Singh, S. et al. A cloud-based two-layer text classification framework for mental health screening with sarcasm and emoji-aware sentiment analysis. Sci Rep (2026). <https://doi.org/10.1038/s41598-026-58453-7>

Journal : Forensic Science International – Reports (Impact Factor: **1.22**) (**Q3** – 47th Percentile)

Mani, R. T., Palimar, V., Pai, M. S., Shwetha, T. S., & Nirmal Krishnan, M. (2025). An evolution of forensic linguistics: From manual analysis to machine learning – A narrative review. Forensic Science International: Reports, 11, 100417. <https://doi.org/10.1016/J.FSIR.2025.100417>

INTELLECTUAL PROPERTY RIGHTS: PATENTS AND COPYRIGHTS

UTILITY PATENTS:

1. **Title of the work:** Sentiment Assessment System and Method

Application Number: 202541055583
Current Status: **Patent Published & RQ Filed**
Date: 13.06.2025

2. **Title of the work:** Multilingual Artificial Intelligence (AI) System and Method for Mental Health Assessment and Support.

Application Number: 202541076810
Current Status: **Patent Published & RQ Filed**
Date: 05.09.2025

3. **Title of the work:** System and Method for Classifying a Mental Health Condition based on Multimodal Data

Application Number: 202641052630
Current Status: **Filed & Patent Published**
Date: 12.06.2026

DESIGN PATENT:

1. **Title of the work:** Sentiment Assessment System and Method

Application Number: 501599-001
Current Status: **Filed & awaiting for Technical Examination**
Date: 11.05.2025

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1. **Title of the work:** Crime Scene Protocol Module: Forensic Science Internship Reference Guide

Certificate Number: LD - 20250172312
Current Status: **Registered**
Date: 22.08.2025

2. **Title of the work:** Pattern Analysis for Compassionate Mental-health Assistance via Narratives (PACMAN): Phase 1: Initialization

Certificate Number: LD - 20250171014

Current Status: **Registered**

Date: 12.08.2025

3. **Title of the work:** Pattern Analysis for Compassionate Mental-health Assistance via Narratives (PACMAN): Phase 2: Model Selection and Prediction

Certificate Number: LD - 20250172524

Current Status: **Registered**

Date: 27.08.2025

4. **Title of the work:** Pattern Analysis for Compassionate Mental-health Assistance via Narratives (PACMAN): Phase 3: Deliver Findings

Certificate Number: LD - 20250172524

Current Status: **Registered**

Date: 10.10.2025

5. **Title of the work:** Website Architecture and User Flow for An Emotional Well-Being Platform-Studentsphere : User and Admin Authentication Flow

Certificate Number: LD - 20250176727

Current Status: **Registered**

Date: 03/11/2025

6. **Title of the work:** Website Architecture and User Flow for An Emotional Well-Being Platform-Studentsphere : Overall Website Structure and Features.

Certificate Number: LD - 20250176052

Current Status: **Registered**

Date: 21/10/2025

7. **Title of the work:** Website Architecture and User Flow for An Emotional Well-Being Platform-Studentsphere : Sentiment Analysis and Data Processing.

Certificate Number: LD - 20250174262

Current Status: **Registered**

Date: 23/09/2025

8. **Title of the work:** Sentiment analysis and mental health assessment using Azure tools: a comparative study with pre-existing models for enhanced detection and classification - Overview architecture diagram.

Certificate Number: LD - 20250173454
Current Status: **Registered**
Date: 09/09/2025

9. **Title of the work:** Sentiment analysis and mental health assessment using Azure tools: a comparative study with pre-existing models for enhanced detection and classification - Detailed architecture diagram.

Certificate Number: LD - 20250174267
Current Status: **Registered**
Date: 23/09/2025

GRANT WRITING

- ✚ Leveraged grant writing skills to secure funding from diverse organizations, including the Department of Health Research (DHR), the Indian Council of Medical Research (ICMR), and the Cognitive Science Research Initiative (CSRI).

ACADEMIC EXCELLENCE

- ✚ Demonstrated a strong commitment to academic excellence by applying for prestigious fellowships, such as the **Commonwealth Split-Site Ph.D. Scholarship** in 2023. This ongoing pursuit reflects my dedication to securing advanced academic opportunities.

AWARDS & ACCOLADES

- ✚ **Ekalavya – Ignite Travel Grant Awardee** Issued by Ignite Life Science Foundation · Aug 2025 : The travel grant is for travel by young scientists to high quality international meetings to present their work to their peers from the global science community.
- ✚ “Best Volunteer Award for Community Services” – School of Service Learning, Loyola College, Chennai – 34.
- ✚ Served as the “Nominated Quality Assurance – Springboard Member” (Internal Quality Assurance Cell – IQAC) at Loyola College.
- ✚ Received an offer to attend the SCCS Pre – Conference Module cum Main Conference Conducted by the Indian Institute of Science (IISc) and Indian Academy of Science [IAS]. As one of the Top 16 Students All over India.

WORKSHOPS

- ✚ On March 16-17, 2023, I participated in the **Industry-Academia Collaboration Workshop on Methodological Approaches to Evidence Synthesis (MAES)**. This workshop was a joint initiative by **Pfizer Ltd., India, Kasturba Medical College (Mangalore & Manipal), and Manipal College of Health Professionals (MCHP)**.
- ✚ Completed a **workshop on the International Council for Harmonization of Good Clinical Practice (ICH-GCP) principles and regulations**. The workshop, organized by the **Centre for Bioethics, MAHE**, and Manipal in collaboration with Manipal - GOK Incubator and Forum for Ethics Committees Review in India (FERCI), took place on March 25, 2023.
- ✚ I attended the **Wildlife Forensic Bootcamp** conducted by the **National Forensic Science University, Goa Campus**, collaborating with **Thermofisher Scientific, Eppendorf, and aquatic life sciences** on May 14th and 15th, 2023. Grateful for the fantastic experience, I appreciate the organizers, speakers, and fellow participants who fostered an environment of learning, growth, and camaraderie. **Dr. Naveen Chaudhary**, Campus Director, shaped my dedication to wildlife conservation.
- ✚ Attended the **ICMR-funded "Capacity Building Workshop on Inclusive Healthcare"** workshop as a delegate on May 30th, 2023, at **Manipal College of Nursing, MAHE**. This workshop explored strategies, trends, and best practices for promoting inclusivity and equity in healthcare delivery. The workshop significantly impacted my research focus, providing valuable insights that will contribute to developing inclusive healthcare policies and practices.
- ✚ I actively participated in the **three-day National Level Workshop on "The Construction of Research Instruments in Social Sciences: Validation and Development (Hands-on Fusion; Bridging Theory and Practice)"** held September 14-16, 2023. Organized by the **Yenepoya-Research Centre for Finance and Entrepreneurship Development (YEN-REFINED)** at Yenepoya Institute of Arts, Science, Commerce, and Management (YIASCM), Yenepoya (Deemed to be University), Mangalore.
- ✚ I actively participated in the **pre-conference workshop** titled **"Gender-Based Vulnerabilities at Household Level,"** held on February 7, 2024, at Kasturba Medical College, Mangalore. This workshop, part of **IAPSMCON 2024** and organized by the Department of Community Medicine, explored the complexities of gender-based vulnerabilities within households, their impact on health, and the critical role of healthcare professionals.
- ✚ **One Day Hands-on Ballistics Workshop on "Analysis of Gun Shot Residues"** Conducted by the Department of Forensic Medicine, Kasturba Medical College, Manipal.
- ✚ **Two hands-on workshops on Forensic Ballistics and Forensic Toxicology**, conducted by Mr. BC Ravindra (Asst. Dir. Retd), FSL Bangalore, and Mr. VG Nayak (Dy Dir. Retd), FSL Bangalore, organized by CCIF and Department of Forensic Medicine and Toxicology, KMC, Manipal.

- ✚ **Cyber Forensics – Hands-on Workshop** conducted by Mr. Samarth Bhaskar Bhat (Reverse Engineering Infosec, Bangalore), organized by the Department of Forensic Medicine and Toxicology, KMC, Manipal.
- ✚ Cyber Security Workshop, organized by the **Centre for Clinical and Innovative Forensics**, KMC, Manipal, and conducted by Mr. Ananth Prabu.
- ✚ Attended a **State Level Workshop** on “**Techniques in Plant Biology and Biotechnology**” organized by Tamil Nadu Science City in collaboration with the Centre for Advanced Study in Botany – University of Madras.
- ✚ Participated in the **pre-conference workshop** on “**Evidence Synthesis and Its Application in Adolescent Mental Health Research**” at **AMHCON-2024**, organized by Prasanna School of Public Health, MAHE, Manipal, November 27, 2024.
- ✚ Participated in the **pre-conference workshop** on “**Creating a Safe Space for Children: School Level Intervention Through Suicide Prevention Gatekeeper Training**” at **AMHCON-2024**, organized by Prasanna School of Public Health, MAHE, Manipal, on November 27, 2024.
- ✚ Completed a **6-hour online training workshop** on “**LGBTQIA+ Affirmative Healthcare**” as part of the **iHEAR RainbowMed project** on December 17, 18, and 19, 2024.
- ✚ Participated in the “**Patent Drafting for Medical Devices**” **workshop** organized by **Medical Instruments, Devices and Allied Services (MIDAS) DST – MAHE HUB**, MAHE, Manipal, on November 27, 2024.

CERTIFICATE COURSE ACCOMPLISHMENTS

- ✚ Completed **Phase I of the Leadership in Mental Health Course (Online)** from **September 13, 2024, to November 30, 2024, engaging in 66 contact hours of lectures and panel discussions**. This globally recognized capacity-building program, organized by **Sangath India**, featured expert-led sessions covering critical topics in global mental health, mental health systems strengthening, community-based interventions, and policy advocacy. The course provided insights into evidence-based practices, mental health leadership strategies, and cross-cultural perspectives in mental health care delivery. The program emphasized **multidisciplinary approaches to mental health challenges, integrating research, clinical expertise, and policy frameworks** to equip participants with practical skills applicable in diverse mental health settings.
- ✚ Completed the certificate course “**Certificate Course in Scoping Review Methodology**” from 7th to 10th August 2024, organized by the Centre for Evidence-informed Decision-making (CEIDeM), Department of Health Information, Prasanna School of Public Health, Manipal Academy of Higher Education (MAHE), Manipal.

- ✚ Completed the certificate course **"Meta-Analysis and Meta-synthesis"** on 28th – 31st December 2023, organized by the Centre for Toxicovigilance & Drug Safety, Manipal College of Pharmaceutical Science, MAHE, Manipal.
- ✚ Completed the **"Research & Publication Ethics"** course offered by the Centre for Doctoral Studies, Directorate of Research, Manipal Academy of Higher Education (MAHE), Manipal, in July 2023.
- ✚ Completed the certificate course **"Biostatistics, Epidemiology, and Research Methodology"** offered by the Department of Data Science at Manipal Academy of Higher Education (MAHE) in April 2023.
- ✚ Completed the online non-credit course **"Digital Footprint"** on September 9, 2022. This course was offered through Coursera and authorized by the University of Edinburgh.

TRAINING & CONFERENCE

- ✚ Presented a paper at an **International Conference** titled **"JUSTICE REIMAGINED: UNLEASHING AI-POWERED FORENSIC LINGUISTICS IN CLINICAL LEGAL EDUCATION FOR ETHICAL INNOVATION AND GLOBAL ACCESS "** at the **International Journal of Clinical Legal Education 2025** in **Monash Law School, Melbourne, Australia** between **19th to 21st November 2025**.
- ✚ Presented a paper virtually at an **International Conference** titled **" TRANSFORMING RURAL MENTAL HEALTH: UNLEASHING THE POWER OF AI IN CARE DELIVERY "** at the **Rural Health International Place-based Education and Research (RHIPER) Conference 2025** in **Murray Bridge, Australia** between **29th to 31st October 2025**.
- ✚ Presented a poster at an **International Conference** titled **" FORENSIC LINGUISTICS MEETS MACHINE LEARNING: A PSYCHOLINGUISTIC MODEL FOR SUICIDE NOTE AUTHENTICITY ANALYSIS."** at the **"Manipal International Forensic Science Conference 2025"** organized by **Department of Forensic Medicine and Toxicology, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, India**, on **04th & 05th October 2025**.
- ✚ Presented a paper at an International Conference titled **" STUDENTSPHERE: A SENTIMENT-DRIVEN PLATFORM FOR ENHANCING STUDENT MENTAL HEALTH AND WELL-BEING "** at the **Asia Pacific Conference of Adolescent Mental Health (AMHCON) – 2024**, organized by **Prasanna School of Public Health, MAHE, Manipal**, on November 28-29, 2024.
- ✚ Presented a scientific poster titled **" ANALYZING SENTIMENTS, NURTURING MINDS: AZURE TEXT CLASSIFICATION FOR MENTAL HEALTH SCREENING "** at Manipal Quest 1.0, an interdisciplinary conference hosted by Manipal Academy of Higher Education on June 9, 2024.
- ✚ Attended the **Student Conference on Conservation Science (SCCS)** – Bengaluru held virtually between 27th September 2021 and 30th September 2021.

- ✚ Attended a training program on “The Exportation & Health Benefits of Herbal Plants” organized by Enactus – SIFE at Loyola College, Chennai.

RESEARCH INTEREST

- ✚ Forensic Linguistics
- ✚ Crime Scene Investigation
- ✚ Forensic Questioned Documents
- ✚ Cyber Forensics
- ✚ Digital Forensics
- ✚ Cyber Security
- ✚ Forensic Photography

PROFESSIONAL AFFILIATION AND RESPONSIBILITIES

- ✚ Coordinated as an **Organizing Committee Member – Scholar** at the “**Manipal International Forensic Science Conference 2025**” organized by Department of Forensic Medicine and Toxicology, Kasturba Medical College, Manipal, Manipal Academy of Higher Education, India, on 04th & 05th October 2025.
- ✚ Served as a **Web Co-ordinator** for the Manipal Academy of Higher Education (MAHE) – Student Research Forum under the aegis of Directorate of Research (DoR), MAHE, Manipal, since January 2024, and organized several events as an Organizer and Web Co-ordinator at MAHE.
- ✚ Served as an **Organizer** at the “**Interdisciplinary Conference on Healthcare and Technical Research – 2024**”, organized by MAHE-Student Research Forum under the aegis of the Directorate of Research, Manipal Academy of Higher Education, Manipal, India, on 18th & 19th December 2024.
- ✚ Served as an **Organizer** at the “**Manipal Research Colloquium – 2024**”, organized by the MAHE-Student Research Forum under the aegis of the Directorate of Research, Manipal Academy of Higher Education, Manipal, India, on 12th & 13th April 2024.
- ✚ Coordinated as an **Organizing Committee Member – Scholar** at the “**International Conference on Nanoscience and Nanotechnology**” organized by Manipal Academy of Higher Education, India, on 29th February & 1st March 2024.
- ✚ Coordinated as an **Organizing Committee Member** at the ICMR-funded “**Capacity Building Workshop on Inclusive Healthcare**” workshop on May 30th, 2023, at Manipal College of Nursing, MAHE.
- ✚ Volunteered in various departmental activities, such as seminars, guest lectures, departmental events, etc.

- ✚ Served as the **Committee Member** of diverse committees in the Department Festival Genotsav 2018 & 2019.
- ✚ Volunteered for the Maintenance of the Herbal Garden along with Prof. Antoine Lebel at the Department of Plant Biology & Biotechnology.
- ✚ Rendered guidance to fellow students in the **Mushroom cultivation Laboratory practices**.

SKILLS & COMPETENCIES

- | | |
|---------------------------|-------------------------------------|
| ✚ Time management | ✚ Enthusiastic |
| ✚ Prioritization of tasks | ✚ Motivator |
| ✚ Decision making | ✚ Positive Thinker |
| ✚ Adaptive Learner | ✚ Proficient in Computer Operations |
| ✚ Collaborator | |

PERSONAL DOSSIER

- ✚ Date of Birth: 22nd December, 1999
- ✚ Gender: Male
- ✚ Marital Status: Unmarried
- ✚ Hobbies and Interests: Technology, Gadgets, Poetry, Designing
- ✚ Linguistic Proficiency
 - Tamil (Native Language)
 - English (Read, Speak & Write)
 - Kannada (Speak)

DECLARATION

I hereby declare that the above-given information is accurate and correct to the best of my knowledge.

Date: 26.06.2026

Yours sincerely,

R. Thamizh Mani
(**Thamizh Mani R**)